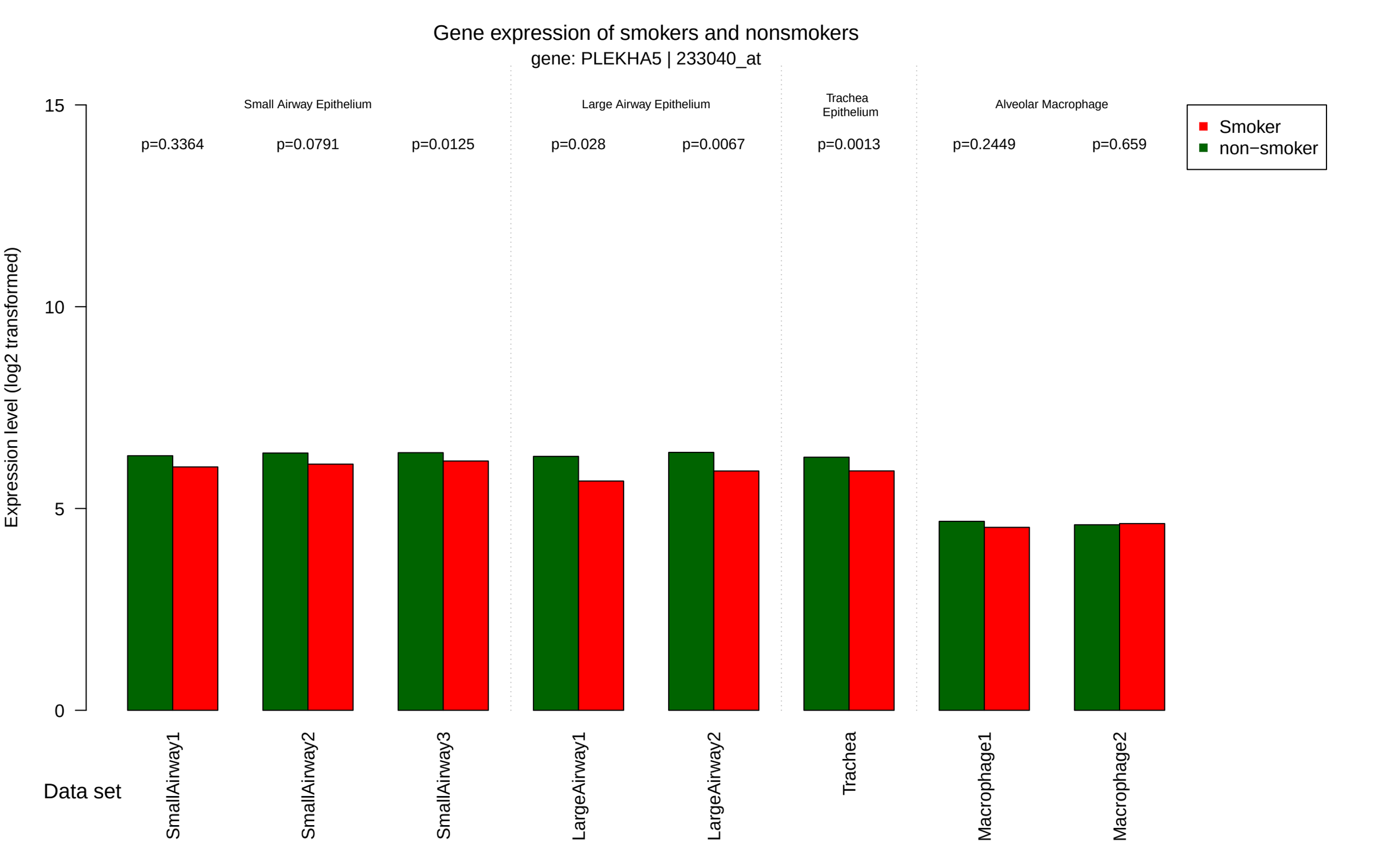
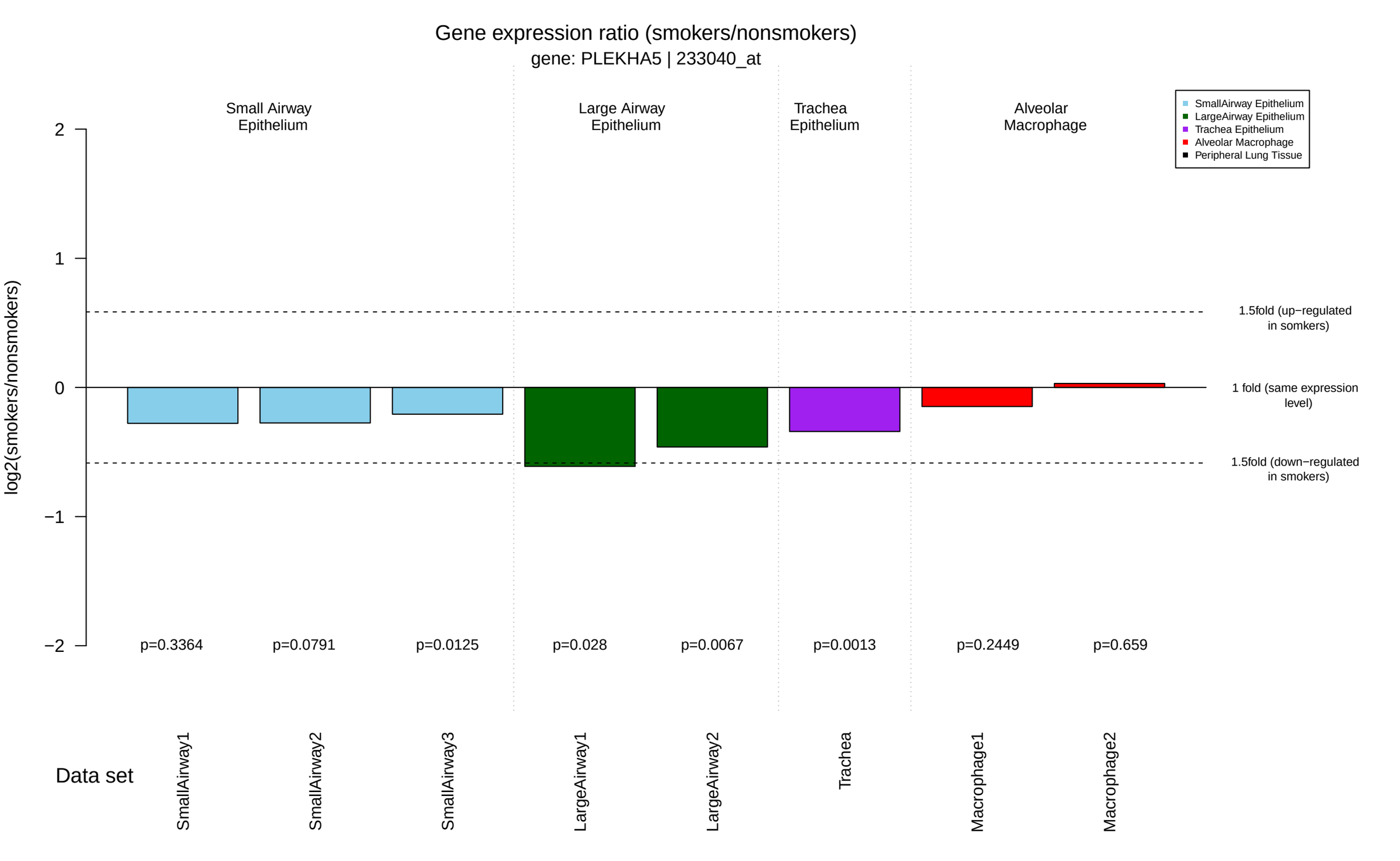
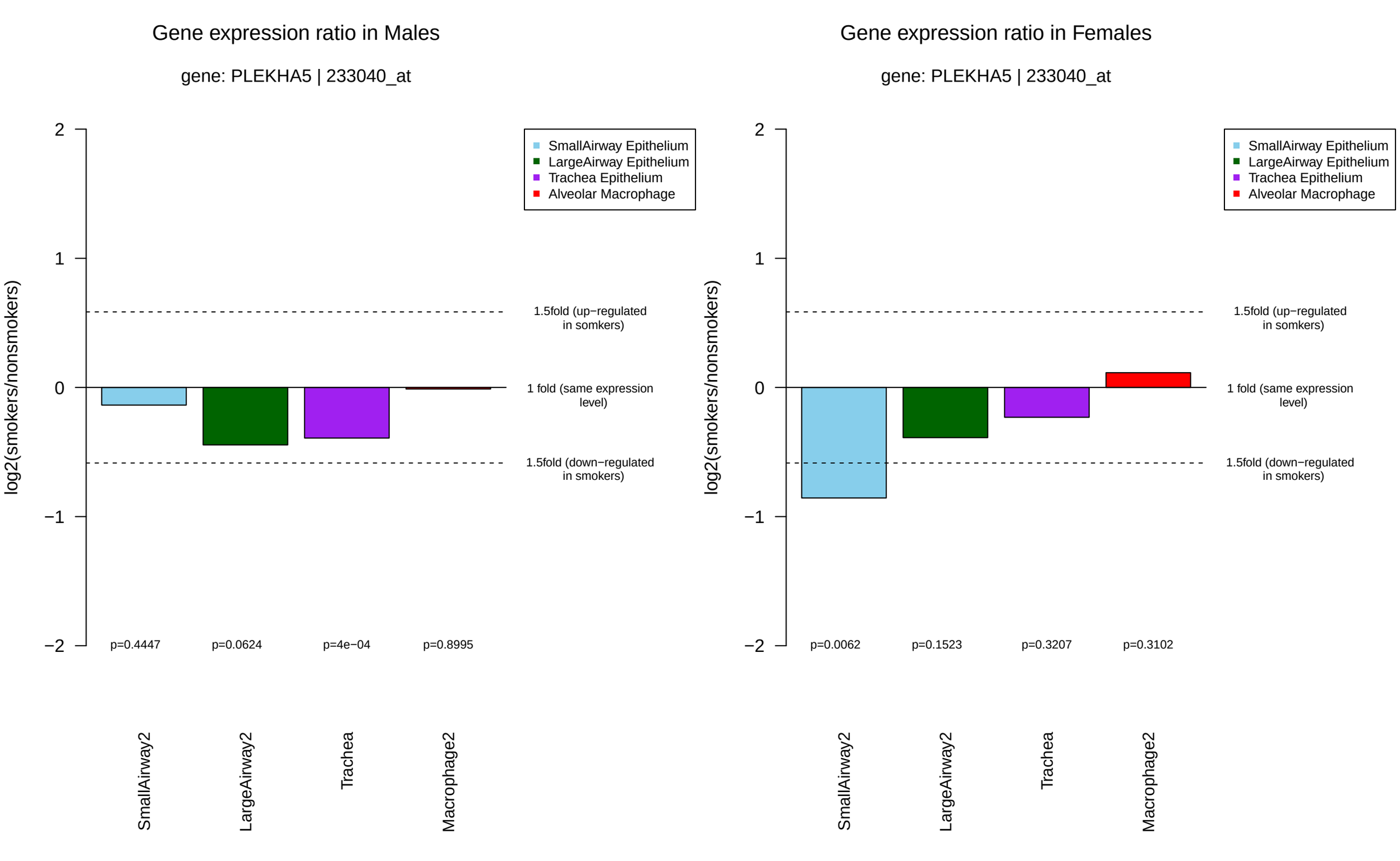
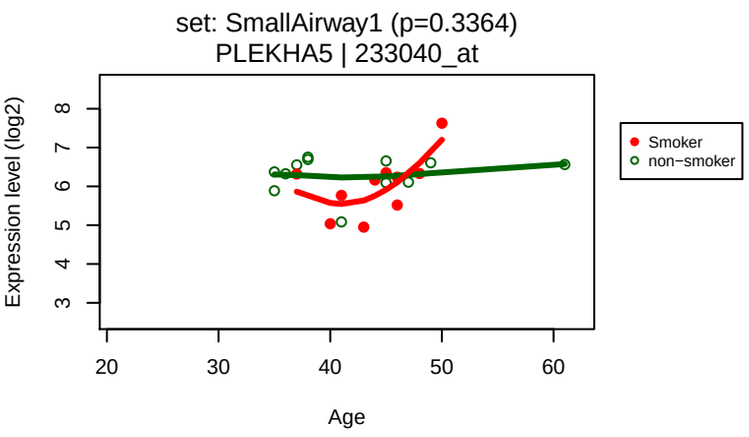
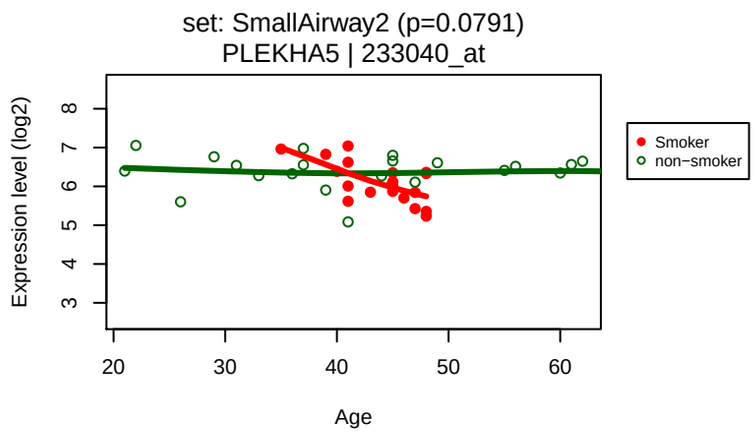
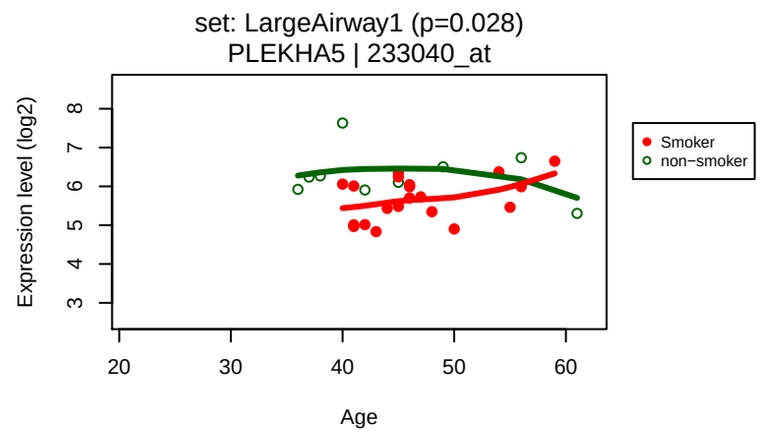

Gene expression of smokers and nonsmokers
gene: PLEKHA5 | 233040_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3364
p=0.0791
p=0.0125
p=0.028
p=0.0067
p=0.0013
p=0.2449
p=0.659
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

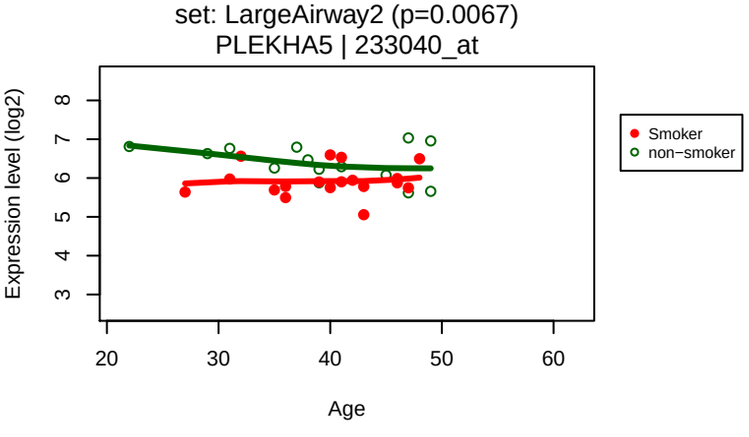

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA5 | 233040_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3364
p=0.0791
p=0.0125
p=0.028
p=0.0067
p=0.0013
p=0.2449
p=0.659
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

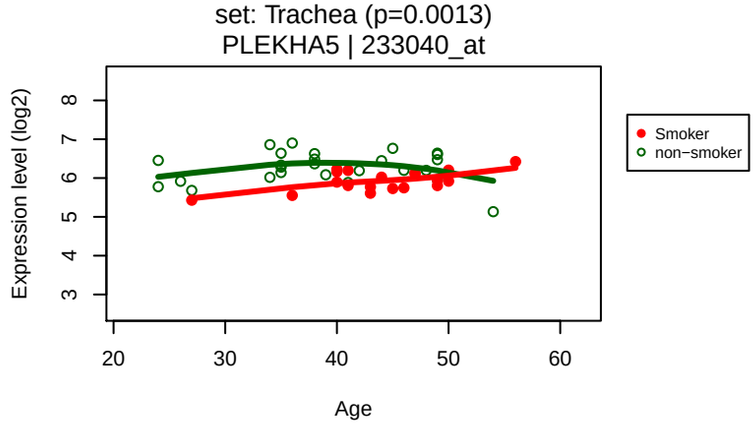

Gene expression ratio in Males
gene: PLEKHA5 | 233040_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4447
p=0.0624
p=4e−04
p=0.8995
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA5 | 233040_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0062
p=0.1523
p=0.3207
p=0.3102
SmallAirway2
LargeAirway2
Trachea
Macrophage2

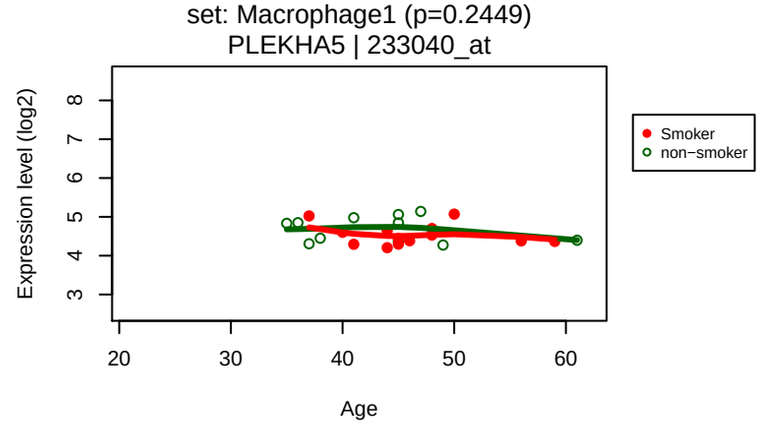

set: SmallAirway1 (p=0.3364)
PLEKHA5 | 233040_at
Expression level (log2)
Age
Smoker
non−smoker

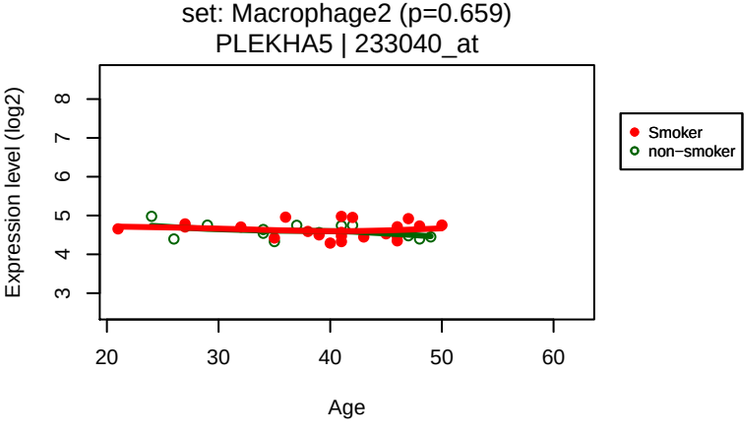

set: SmallAirway2 (p=0.0791)
PLEKHA5 | 233040_at
Expression level (log2)
Age
Smoker
non−smoker

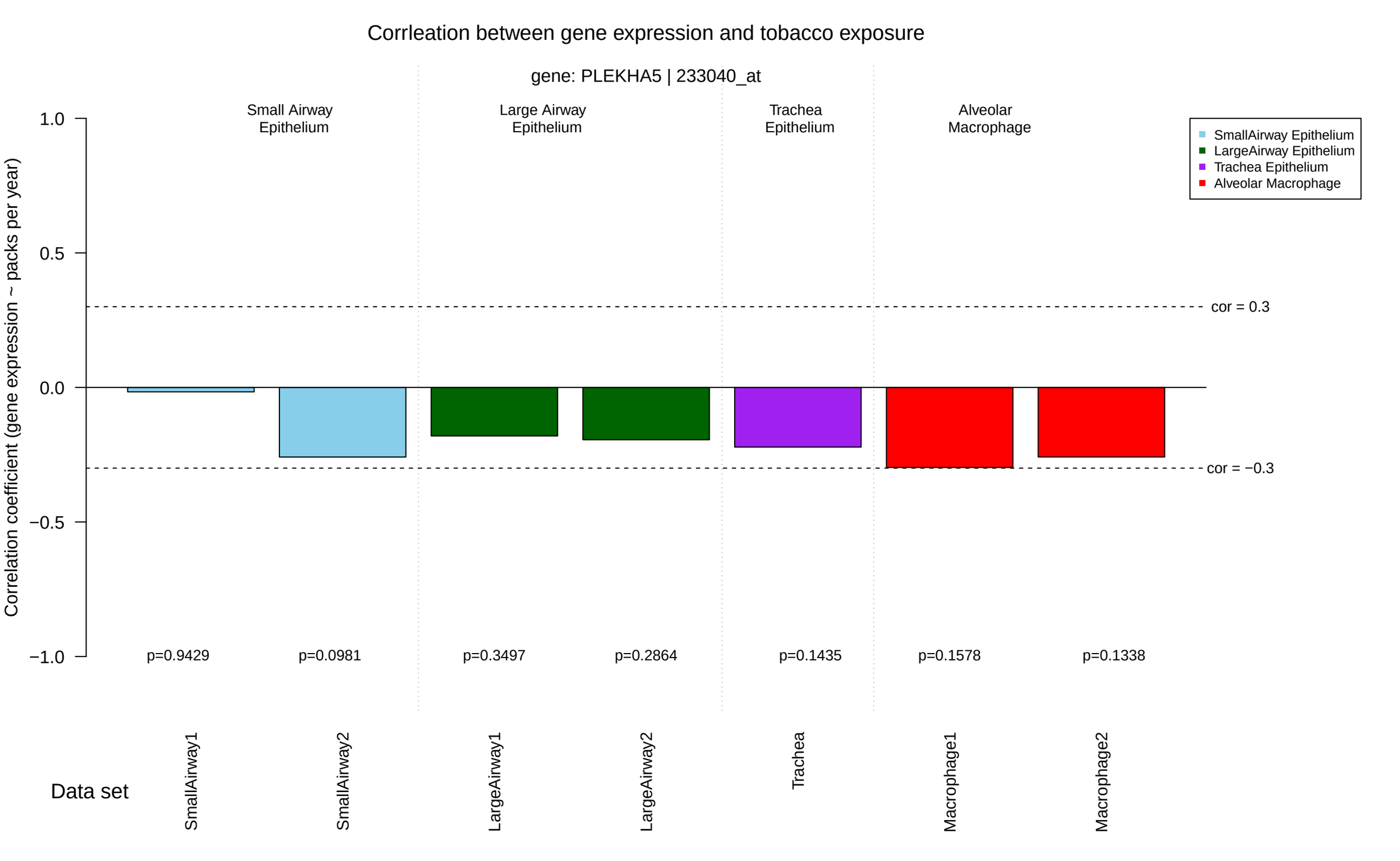

set: LargeAirway1 (p=0.028)
PLEKHA5 | 233040_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0067)
PLEKHA5 | 233040_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
PLEKHA5 | 233040_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2449)
PLEKHA5 | 233040_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.659)
PLEKHA5 | 233040_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA5 | 233040_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9429
p=0.0981
p=0.3497
p=0.2864
p=0.1435
p=0.1578
p=0.1338
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2